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# 07 – Reproducibility Protocol
## Software demo (no MF tree)
``bash
cd STANDALONE_FLAGSHIP && ./RUN_DEMO.sh
``
- Python 3.9+, stdlib only
- Frozen inputs: `specimen/*.json`
- Outputs generated by `engine/runtime.py` into `results/`
- Expect PASS and max gate  $\Delta > 0.05$ 
## Deterministic
IR execution given package + stimuli is deterministic.
## Requires hardware
Live FPGA hot-load fingerprints; physical robot loop.
## Artifacts
Bitstream: `STANDALONE_FLAGSHIP/hardware/rcf_multislot.bin`
Manifest: `hardware_manifest.json`
```